Agentic Multimodal Fatigue Fingerprinting

Comprehensive Technical Reference, User Workflow & Borg RPE Scale Manual

EXECUTIVE SUMMARY & SYSTEM PURPOSE:
The **Agentic Fatigue Fingerprinting Framework** is a physiological monitoring system designed for researchers, ergonomic specialists, and clinicians. Grounded on the benchmark **WSD4FEDSRM** dataset (34 participants), the system quantifies neuromuscular and cardiorespiratory fatigue in real-time, predicts exertion on the standardized **Borg RPE Scale (6–20)**, translates attributions via SHAP, and deploys 4 autonomous AI agents to enforce safety policies.

1. The Fatigue Scale: Borg Rating of Perceived Exertion (RPE 6–20)

Fatigue in this system is quantitatively measured and predicted on the standardized **Borg RPE (Rating of Perceived Exertion) Scale**, ranging from **6 to 20**. Developed by Gunnar Borg, this scale correlates directly with physiological exertion and heart rate (RPE × 10 ≈ Heart Rate in BPM for healthy adults).

Borg RPE Value	Exertion Level	Physiological & Muscular State	System Safety Action
6	No Exertion At All	Complete rest. HR baseline (~60 BPM), normal EMG frequency spectrum, high HRV RMSSD.	**SAFE** (Normal Baseline)
7 – 10	Extremely Light / Very Light	Minimal muscle engagement. Regular breathing and stable heart rate.	**SAFE** (Optimal Load)
11 – 12	Light Exertion	Comfortable working load. Slight increase in sEMG RMS amplitude.	**SAFE** (Target Baseline)
13 – 14	Somewhat Hard	Early fatigue onset. sEMG Mean Power Frequency (MPF) starts dropping, HR elevates slightly.	**MONITOR** (Moderate Exertion)
15 – 16	Hard / Heavy Exertion	Significant muscular fatigue. MPF/MDF spectral drop >15%, HRV RMSSD declines, respiration elevates.	**WARNING THRESHOLD** Sentinel Agent triggers yellow alert.
17 – 18	Very Hard Exertion	Severe muscle exhaustion. High motor unit recruitment, micro-tremor arm jitter on IMU, elevated EEG Alpha/Beta.	**CRITICAL RISK** Sentinel Agent orders task pause.
19 – 20	Extremely Hard / Maximal	Complete muscular failure and exhaustion. High risk of strain or acute injury.	**EMERGENCY STOP** Mandatory rest required.

2. How a User Uses the System (Step-by-Step Workflow)

A researcher, ergonomic consultant, or clinician uses the system following this 5-step operational workflow:

Step 1: Open the Interactive Dashboard

Launch the dashboard by navigating to http://127.0.0.1:8000/dashboard/ in any standard browser while the FastAPI server is running.

Step 2: Select Participant & Set Safety Threshold

In the top panel, select a participant (e.g., **Subject P03**) from the dropdown. The system instantly loads their individual physical parameters (age, sex, arm length, MVIC force). Adjust the **RPE Warning Threshold slider** (default: 15.0).

Step 3: Monitor Live Multimodal Signals

In **Tab 1 (Multimodal Signals)**, observe real-time physiological plots:

- *sEMG Envelope:* Watch for power increases or frequency drops across shoulder muscles.
- *ECG & HRV:* Monitor Heart Rate and RMSSD drops signaling cardiorespiratory stress.
- *EEG & Tremor:* Check Alpha/Beta ratio and IMU arm tremor jitter for neuromuscular fatigue.

Step 4: Trigger Multi-Agent AI Analysis

Click the **'Trigger Agent Check'** button. The 4 AI Agents run sequentially:

- **Data Agent** verifies signal validity.
- **Personalization Agent** calibrates baseline offsets.
- **Explainability Agent** extracts top SHAP drivers.
- **Safety Sentinel Agent** evaluates risk against threshold and emits instant warnings.

Step 5: Inspect SHAP Drivers & Agent Logs

Switch to **Tab 2 (SHAP Attributions)** to see exact feature impact rankings (e.g., why fatigue rose) and **Tab 4 (Agent Activity Log)** to read plain-language medical narratives generated by the agents.

3. Dataset & Measurements Used

Grounding data comes from the **WSD4FEDSRM** dataset (34 subjects, P01 to P34) during shoulder movement tasks across 6 load conditions (30%, 50%, 70% MVIC).

Modality	Sensors / Channels	Extracted Features & Measurements	Physiological Fatigue Significance
sEMG (Electromyography)	6 Shoulder Muscle Channels: • Ant. Deltoid • Infraspinatus • Lat. Dorsi • Pect. Major • Post. Deltoid • Upp. Trapezius	• Root Mean Square (RMS) • Mean Power Frequency (MPF) • Median Frequency (MDF) • Waveform Length (WL) • Zero Crossings (ZC)	Fatigue causes MPF/MDF shift to lower frequencies and RMS amplitude rise due to motor unit synchronization.
ECG & Respiration	• Lead-II ECG • Chest Respiration Band	• Heart Rate (HR) • HRV: SDNN, RMSSD, pNN50 • LF/HF Ratio • Respiration Rate (BrPM)	Autonomic strain lowers HRV (RMSSD drop) and increases heart and breathing rates as fatigue progresses.
EEG & IMU	• Motor/Prefrontal EEG • 3-Axis Arm Acceleration & Gyro	• Delta, Theta, Alpha, Beta, Gamma power • Alpha/Beta Power Ratio • 3-Axis Tremor RMS Jitter	Central fatigue increases Alpha & Theta power, while motor unit fatigue causes physical micro-tremor arm jitter.

4. Multi-Agent AI System Architecture

Agent Name	Core Function & Responsibility	Output / Narrative Generated
■■ **Data Intelligence Agent**	Audits raw physiological signals, calculates SNR, checks channel continuity, flags motion artifacts.	*'Signal Audit Certified: All 6 EMG channels, ECG, and IMU streams intact.'*
■ **Personalization Agent**	Calculates subject-specific physical baseline offsets based on P01–P34 parameters (arm length, MVIC force).	*'Personalization Applied for P03: Population baseline adjusted by physical calibration.'*
■ **Explainability Agent**	Translates SHAP attribution vectors into plain-language physiological narratives detailing top fatigue drivers.	*'Primary Driver: EMG MPF drop (-1.2 SHAP). Secondary: Heart rate (+0.85 SHAP).'*
■ **Decision & Safety Sentinel**	Enforces RPE safety thresholds (RPE > 15 warning, RPE > 18 critical), detects over-exertion, and commands stop.	*'SAFE: Current predicted RPE 12.4 is within safety threshold (<15.0).'*

5. Verification & REST API Reference

REST Endpoints: Server running on http://127.0.0.1:8000 serving /api/health, /api/subjects, /api/signals/stream, and /api/agents/analyze.

Automated Test Suite: 285 out of 285 unit tests passing across all 12 modules (100% test pass rate).